CRYPTO SUPER 40 QUE ANS (GTU)

Q1. Define Cryptography. Explain types of Cryptography

Definition of Cryptography

Cryptography is a method of keeping information safe.
It changes normal readable data (plaintext) into unreadable data (ciphertext).
Only the person who has the correct key can read the secret message again.
Its main purpose is to protect data from hackers and keep it private.

Types of Cryptography

1.Symmetric-Key Cryptography
Same key is used to lock (encrypt) and unlock (decrypt) the message.
It is fast and used for large data.
Example: AES, DES.

2. Asymmetric-Key Cryptography

Uses two different keys:

Public key → used to encrypt the message
Private key → used to decrypt the message
It gives more security.
Example: RSA.

3. Hash Functions (Key-less Cryptography)

No key is used.
It converts data into a fixed-size code called a hash.
It cannot be reversed back to original data.
Used to check integrity.
Example: SHA-256, MD5.

Q2 (A): What is Computer Security?

Computer security means protecting a computer system from unauthorized use or attacks.
It keeps data safe from hackers, viruses, and harmful software.
It ensures that only the correct users can access the computer and its information.
It protects the computer's hardware, software, files, and network.

Its main goals are:

Confidentiality – keeping data secret
Integrity – keeping data correct

Availability – making sure systems work when needed

Q2 (B): Explain Information Security

Information security protects information from being accessed or changed by unauthorized people.
It applies to all types of information: digital files, documents, emails, and stored data.
It prevents misuse, loss, or leakage of important data.
It uses tools like passwords, encryption, and access control.

Main goals (CIA):

Confidentiality – information is secret
Integrity – information is accurate
Availability – information is available when needed

Q2 (C): Explain Network Security

Network security protects data while it is being sent through a network (like internet, Wi-Fi).
It stops hackers from entering the network or stealing information.
It protects computers, servers, and communication channels from attacks.
It uses firewalls, antivirus, encryption, and intrusion detection.
It prevents attacks like malware, phishing, DoS attacks, and unauthorized access.

Q3 (A): Define Essential Network and Computer Security Requirements

Confidentiality: Ensures that only authorized users can access data.
Integrity: Ensures that data is not altered or tampered with.
Availability: Ensures that the system and data are available when needed.
Authentication: Verifies the identity of users before allowing access.
Non-repudiation: Ensures that the sender cannot deny sending a message.
Access Control: Restricts unauthorized users from accessing resources.

Q3 (B): Explain Security Basics in Detail

Confidentiality: Keep information secret from unauthorized users.
Integrity: Make sure data is accurate and has not been changed.
Availability: Ensure systems and information are available for use whenever needed.
Authentication: Verify the identity of users, devices, or systems.
Authorization/Access Control: Allow only permitted users to access resources.
Non-repudiation: Ensure that a sender cannot deny sending a message.
Accountability: Track user actions to detect misuse or attacks.

Q4. Explain The OSI Security Architecture in Brief

The OSI Security Architecture defines how to provide security in network communication.
It is based on the OSI model, which has seven layers for data transmission.

The architecture identifies security services like:

Confidentiality: Keep data secret
Integrity: Prevent data modification
Authentication: Verify user or system identity
Access Control: Restrict unauthorized access
Non-repudiation: Prevent denial of actions

It also defines security mechanisms like encryption, digital signatures, and firewalls to protect data.
Helps in designing secure networks and detecting threats at different layers.

Q5. Difference between Active Attack & Passive Attack
1.Active Attack

The attacker modifies, deletes, or adds data in the system or network.
It affects the normal operation of the system.
Examples: Masquerading, Replay attack, Denial-of-Service (DoS).
Active attacks are easily detectable because they change data or behavior.

2. Passive Attack

The attacker only observes or listens to the communication without changing it.
The system continues to work normally.
Examples: Eavesdropping, Traffic analysis.
Passive attacks are harder to detect because data is not altered.

Q6. Define Security Services
Security services are functions that help protect information and systems from threats.

Main Security Services

Authentication: Verifies the identity of users, devices, or systems before access.
Access Control: Ensures only authorized users can access resources.
Data Confidentiality: Keeps information secret from unauthorized users.
Data Integrity: Ensures that data is not altered or tampered with.
Non-repudiation: Prevents a sender from denying that they sent a message.
Availability: Ensures that systems and data are available whenever needed.

Q7. Explain Security Mechanisms – Specific & Pervasive Security Mechanisms
1.Specific Security Mechanisms

These are designed to provide a particular security service.

Examples include:

Encryption: Protects data confidentiality.
Digital Signatures: Ensure authentication and non-repudiation.
Access Control Lists (ACLs): Restrict access to authorized users.
Hash Functions: Ensure data integrity.

2. Pervasive Security Mechanisms

These are used to provide security across multiple services or systems.

Examples include:

Firewalls: Protect networks from external threats.
Audit Logs: Track activities for accountability.
Intrusion Detection Systems (IDS): Detect suspicious activities.
Security Policies: Define rules and guidelines for all systems.

Q8. Explain Model for Network Security

A network security model shows how to protect data during transmission between sender and receiver.

Components of the model:

Sender: Sends the original message (plaintext).
Encryption Algorithm: Converts plaintext into ciphertext to protect it.
Key: Used for encryption and decryption.
Communication Channel: The network where data travels.
Receiver: Receives ciphertext and converts it back to plaintext using the key.

The model also considers attackers, who may try to intercept, modify, or steal data.
Security mechanisms like encryption, digital signatures, and authentication are used to secure communication.

Q9. Explain Key-less, Single-key, and Two-Key Algorithms

1. Key-less Algorithms

Do not use any key for encryption or decryption.
Usually used for hashing to check data integrity.
Example: SHA, MD5.

2. Single-key Algorithms (Symmetric Key)

Use the same key for both encryption and decryption.
Fast and suitable for large data.
Example: AES, DES.

3. Two-key Algorithms (Asymmetric Key)

Use two different keys:

Public key → for encryption
Private key → for decryption

Provides higher security but slower than symmetric key.
Example: RSA, ECC.

Q10. What is Divisibility and Division Algorithm

1. Divisibility
A number a divides another number b if there exists an integer k such that $b = a \times k$.
Example: 3 divides 12 because $12 = 3 \times 4$.

2. Division Algorithm
For any integers a and b ($b > 0$), we can write:
$a = b \times q + r$, where
q is the quotient
r is the remainder
$0 \leq r < b$

Example: Divide 17 by 5 → $17 = 5 \times 3 + 2$, so $q = 3$, $r = 2$.

Q11. Define Modulo Operator & Modular Arithmetic Properties

1. Modulo Operator

The modulo operator (mod) gives the remainder after division.
Example: 17 mod 5 = 2, because 17 ÷ 5 leaves remainder 2.

2. Modular Arithmetic Properties

Addition: $(a + b) \bmod n = ((a \bmod n) + (b \bmod n)) \bmod n$
Subtraction: $(a - b) \bmod n = ((a \bmod n) - (b \bmod n)) \bmod n$
Multiplication: $(a \times b) \bmod n = ((a \bmod n) \times (b \bmod n)) \bmod n$

These rules are used in cryptography to perform calculations under modulo.

Q12. What is (Set of Residues mod n)?

is called the set of residues modulo n.
It consists of all integers from 0 to n−1. These are the possible remainders when any integer is divided by n.
Every integer, when divided by n, will leave a remainder that belongs to .
Arithmetic in is done mod n, meaning after each operation (addition, subtraction, multiplication), we take the remainder after dividing by n.
Addition in : (a + b) mod n gives a result in .
Subtraction in : (a − b) mod n is also in .
Multiplication in : (a × b) mod n remains in Example: If n = 5, .

7 mod 5 = 2 → 2 belongs to
12 mod 5 = 2 → 2 belongs to

is very important in cryptography, especially for modular arithmetic, key generation, and encryption/decryption algorithms.
Using , we can easily work with numbers in finite sets, which is necessary for secure and efficient computation in cryptography.

Q13. What is Inverse?
In mathematics and cryptography, an inverse of a number is another number which, when combined with the original number using a certain operation, gives a result of 1.
In modular arithmetic, the inverse of a number a modulo n is a number b such that:

(a \times b) \mod n = 1

The inverse exists only if a and n are coprime, i.e., gcd(a, n) = 1.

Example: For n = 7, the inverse of 3 is 5 because:

3 \times 5 = 15, \quad 15 \mod 7 = 1

Multiplicative inverse is especially important in algorithms like RSA and Affine cipher, where division is done under modulo.

If a number does not have an inverse modulo n, it cannot be used for certain encryption operations.

Finding the inverse can be done using the Extended Euclidean Algorithm, which calculates the value of b efficiently.

Q15. Explain the set of multiplicative inverse in In (set of residues modulo N), a multiplicative inverse of a number a is a number b such that:

(a \times b) \mod N = 1

Only numbers that are coprime with N (gcd(a, N) = 1) have a multiplicative inverse.

Set of multiplicative inverses contains all numbers in that have such an inverse.

Example: For N = 7,

1 × 1 mod 7 = 1 → 1 is its own inverse
2 × 4 mod 7 = 1 → 2 and 4 are inverses
3 × 5 mod 7 = 1 → 3 and 5 are inverses
6 × 6 mod 7 = 1 → 6 is its own inverse

Numbers like 0 do not have a multiplicative inverse because they cannot produce 1.
Multiplicative inverses are important in cryptography, especially in RSA, Affine, and Hill ciphers, to compute keys and decrypt messages.

Q16. Do the Calculation of Addition & Multiplication Tables with Inverse
In modular arithmetic, addition and multiplication tables are created using numbers from .

Addition Table:

Add each pair of numbers and take mod N for the result.

Example: For N = 5,

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Multiplication Table:

Multiply each pair of numbers and take mod N for the result.

Example: For N = 5,

×	0	1	2	3	4
0	0	0	0	0	0

1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Multiplicative Inverse:

From the multiplication table, a number a has an inverse b if (a × b) mod N = 1.

Example for N = 5:

1 × 1 = 1 → inverse of 1 is 1
2 × 3 = 1 → inverse of 2 is 3
3 × 2 = 1 → inverse of 3 is 2
4 × 4 = 1 → inverse of 4 is 4

These tables are useful in cryptography for encryption, decryption, and key generation.

Q17. Explain Euclidean Algorithm for GCD (Greatest Common Divisor)

The Euclidean Algorithm is a method to find the greatest common divisor (GCD) of two integers. GCD of two numbers is the largest number that divides both numbers without leaving a remainder.

Steps of Euclidean Algorithm:

Step 1: Take two numbers, say a and b, where a > b.
Step 2: Divide a by b to get the quotient q and remainder r:

a = b \times q + r

Step 4: Repeat the process until remainder r = 0.
Step 5: The last non-zero remainder is the GCD(a, b).

Example: Find GCD of 56 and 24

56 ÷ 24 = 2 remainder 8 → a = 24, b = 8
24 ÷ 8 = 3 remainder 0 → remainder = 0 → GCD = 8

Importance:

Used in cryptography, number theory, and modular arithmetic.
Helps in finding multiplicative inverses in modular systems like RSA.

Q18. What is Extended Euclidean Algorithm for Multiplicative Inverse

The Extended Euclidean Algorithm is used to find the multiplicative inverse of a number modulo n.
It is an extension of the Euclidean Algorithm, which finds the GCD of two numbers.
Goal: Find integers x and y such that:

a \times x + n \times y = \text{GCD}(a, n)

Steps:

1. Apply the Euclidean Algorithm to find GCD(a, n).
2. Express the GCD as a linear combination of a and n.
3. Solve for x (mod n).
4. x is the multiplicative inverse of a modulo n.

Example: Find inverse of 3 modulo 7

1. Apply Euclidean Algorithm:

7 = 3 × 2 + 1 → remainder 1

2. Express 1 as a combination of 3 and 7:

1 = 7 − 3 × 2
1 = (−2 × 3) + 1 × 7

3. Take x = −2 → modulo 7 → x = 5

4. Answer: 5 is the multiplicative inverse of 3 modulo 7

5. Importance:

Used in RSA, Affine, and other ciphers for decryption and key generation.
Allows solving modular equations efficiently.

Q19. Explain Symmetric-Key Encryption
Symmetric-key encryption is a method of encrypting data where the same key is used for both encryption and decryption.
It is also called secret-key encryption.

Process:

Sender uses the key to encrypt plaintext → produces ciphertext.
Receiver uses the same key to decrypt ciphertext → recovers plaintext.

Features:

Fast and efficient for large amounts of data.
Requires secure key sharing between sender and receiver.

Advantages:

Simple and fast
Good for bulk encryption

Disadvantages:

Key must be kept secret; if compromised, security fails.
Not suitable for communication between many users because each pair needs a unique key.

Examples: AES, DES, 3DES, Blowfish.

Applications:

Encrypting files, emails, messages, and network communication.
Used in VPNs, secure storage, and database encryption.

Q20. Explain Symmetric Cipher Model with Diagram

1.Symmetric Cipher Model Overview
Symmetric cipher uses the same key for encryption and decryption.
The process ensures confidentiality, meaning only authorized users can read the data.

2. Components of Symmetric Cipher Model
Plaintext: Original readable message.
Encryption Algorithm: Mathematical function that converts plaintext into ciphertext using the key.
Key: Secret value shared between sender and receiver.
Ciphertext: Encrypted message, unreadable without the key.
Decryption Algorithm: Converts ciphertext back to plaintext using the same key.

Steps of Symmetric Encryption

1. Sender inputs plaintext and key into the encryption algorithm.
2. Encryption algorithm produces ciphertext.
3. Ciphertext is sent through a network to the receiver.
4. Receiver applies the decryption algorithm using the same key.
5. Original plaintext is recovered.

Diagram Description

Plaintext → [Encryption Algorithm + Key] → Ciphertext → [Decryption Algorithm + Key] → Plaintext

Advantages

Fast and efficient for large data.
Simple implementation.

Disadvantages

Key must remain secret.
Difficult to manage keys for multiple users.

Examples
DES, AES, 3DES, Blowfish.

Q21. Explain Types of Attacks
Attacks are attempts by unauthorized users (hackers) to access, modify, or destroy data or systems.

1. Passive Attacks

Attacker only observes data without altering it.
The system works normally, so it is hard to detect.
Examples:

Eavesdropping: Listening to network communication.
Traffic Analysis: Monitoring patterns of communication.
Goal: Steal information without being noticed.

2. Active Attacks

Attacker modifies, deletes, or inserts data.
Easier to detect because the system behavior changes.

Examples:

Masquerading: Pretending to be someone else.
Replay Attack: Resending captured data to trick the system.
Denial-of-Service (DoS): Making a service unavailable.

3. Other Types of Attacks

1. Man-in-the-Middle (MITM): Attacker intercepts and alters communication between two parties.
2. Phishing Attack: Tricking users into giving sensitive information.
3. Brute Force Attack: Trying all possible passwords to gain access.
4. Malware Attack: Using viruses, worms, or Trojans to damage or steal data.

Q22. Explain Substitution Ciphers

Substitution ciphers are a type of encryption where each letter or symbol in the plaintext is replaced by another letter or symbol.

How It Works

Plaintext letters are replaced using a rule, key, or algorithm.
The receiver uses the key to reverse the process and recover the original message.
The pattern of substitution can be fixed or change during encryption.

Types of Substitution Ciphers

1. Monoalphabetic Cipher:

Uses a single substitution alphabet for the entire message.
Example: Caesar cipher (shift each letter by a fixed number).

2. Polyalphabetic Cipher:

Uses multiple substitution alphabets to make encryption stronger.
Example: Vigenère cipher, Hill cipher.

Features

1. Simple to understand and implement.
2. Provides confidentiality of data.
3. Vulnerable to frequency analysis if used alone (especially monoalphabetic).

Example

Plaintext: HELLO

Using a key: Shift by 3 (Caesar Cipher) → Ciphertext: KHOOR

Importance

Used in historical encryption methods.
Basis for many modern cryptography techniques.

Q23. Explain Types of Monoalphabetic Ciphers Monoalphabetic ciphers use a single substitution alphabet to encrypt the entire message. There are several types:

1. Additive (Caesar) Cipher

1. Each letter of plaintext is shifted by a fixed number.

2. Example: Shift by 3 → A → D, B → E, etc.

3. Encryption formula:

C = (P + K) \mod 26

4. Decryption formula:

P = (C - K) \mod 26

2. Multiplicative Cipher

1. Each letter is multiplied by a key modulo 26.

2. Works only if key and 26 are coprime.

3. Encryption:

C = (P \times K) \mod 26

3. Affine Cipher

1. Combines Additive and Multiplicative cipher.

2. Formula:

C = (P \times K_1 + K_2) \mod 26

4. Decryption:

P = K_1^{-1} \times (C - K_2) \mod 26

4. Monoalphabetic Substitution Cipher

1. Each plaintext letter is replaced with another fixed letter based on a key.

2. Example key:

Plaintext alphabet: A B C D E …

Cipher alphabet: Q W E R T …

3. Stronger than Caesar but vulnerable to frequency analysis.

Features

1. Easy to implement.
2. Security is low for short messages.
3. Used mainly for learning purposes and historical encryption.

Q24. Explain Types of Polyalphabetic Ciphers Polyalphabetic ciphers use multiple substitution alphabets to encrypt plaintext. This makes them more secure than monoalphabetic ciphers.

1. Autokey Cipher

Uses a key that starts with a keyword and then continues with the plaintext itself.
Each letter of the plaintext is shifted by the corresponding key letter.
Example:
Key: SECRET
Plaintext: HELLO → Ciphertext depends on key letters.
Harder to break than simple Caesar cipher.

2. Vigenère Cipher

Uses a repeating keyword to shift letters.
Encryption formula:

$C_i = (P_i + K_i) \mod 26$
$P_i = (C_i - K_i) \mod 26$

Provides better security than monoalphabetic ciphers.

3. Playfair Cipher

Encrypts pairs of letters instead of single letters.
Uses a 5×5 matrix key containing letters of the alphabet.

Rules:

Same row: replace each letter with letter to its right
Same column: replace each letter with letter below
Rectangle: swap letters at rectangle corners

Example: Plaintext = HI → Ciphertext depends on matrix position.

4. Hill Cipher

Uses matrix multiplication to encrypt blocks of letters.
Key is an n×n invertible matrix modulo 26.
Encryption: Multiply plaintext vector by key matrix → ciphertext vector.
Decryption: Multiply ciphertext by inverse of key matrix modulo 26.

5. One-Time Pad Cipher

Uses a random key as long as the plaintext.
Each letter is encrypted by adding key modulo 26.
Perfectly secure if key is truly random and used only once.
Example: Plaintext = HELLO, Key = XMCKL → Ciphertext = (H+X)(E+M)…

6. Features

More secure than monoalphabetic ciphers.
Prevents simple frequency analysis.
Used in historical and classical encryption systems.

Q25. What is Transposition Cipher
A transposition cipher is a type of encryption where the letters of the plaintext are rearranged to create ciphertext.
Unlike substitution ciphers, the letters themselves are not changed, only their positions are changed.
The rearrangement follows a specific rule or key known to both sender and receiver.

How It Works

1. Plaintext is written in rows, columns, or a grid.
2. Letters are read in a different order according to the key.
3. Receiver uses the same key to rearrange letters back to plaintext.

Example

Plaintext: HELLO

Key: 3 (number of columns)

Arrange in 3 columns:

H E L
L O

Read column-wise: H L E O L → Ciphertext: HLEOL

Features

1. Does not change letters, only positions.
2. Security depends on key and method of transposition.
3. Can be combined with substitution ciphers for stronger encryption.

Uses

Historical encryption, puzzles, and some classical military ciphers.

Q26. Explain Keyless Transposition – Rail Fence & Rectangular (Columnar) Cipher

1. Keyless Transposition Cipher

In a keyless transposition cipher, letters of plaintext are rearranged without using a key.
The positions of letters are changed according to a fixed pattern or rule.
Example: Rail Fence and Rectangular (Columnar) methods.

2. Rail Fence Cipher
Letters are written in a zig-zag (rail fence) pattern across multiple rows.
Ciphertext is obtained by reading letters row by row.

3. Example:

Plaintext: HELLO WORLD

2 Rails:

H L O W R D
E L O L

Read row-wise → Ciphertext: HLOWRDELLOL
4. Simple and easy to use, but not very secure.

3. Rectangular (Columnar) Transposition Cipher

1. Plaintext is written in rows of a rectangle.
2. Number of columns = fixed.
3. Letters are read column by column to form ciphertext.

4. Example:

Plaintext: HELLO WORLD

Columns = 4

H E L L
O W O R
L D

Read column-wise → Ciphertext: HOL ELW LOD R

5. Easy to implement, but can be broken if message is short.

Features of Keyless Transposition

1. Only rearranges letters, does not change them.
2. Simple encryption without needing a key.
3. Can be combined with other ciphers for stronger security.

Q27. Explain Keyed Transposition Methods
Keyed transposition ciphers rearrange letters of plaintext using a secret key. The key determines the order of encryption.
1. Keyed Columnar Transposition

Plaintext is written in a rectangle with columns.
Columns are rearranged according to the key.
Steps:

Assign numbers to key letters (alphabetical order).
Write plaintext in rows.

Read ciphertext column by column in key order.

Example:

Plaintext: HELLO WORLD

Key: ZEBRA → Column order: 5,2,3,1,4

Ciphertext is obtained by reading columns in this order.

Provides better security than keyless transposition.

2. Key Inversion Method

Letters in plaintext are rearranged according to a numerical key.
The key indicates the new positions of letters.
Example:

Plaintext: HELLO

Key: 3 1 4 5 2 → Move letters to positions 3,1,4,5,2

Ciphertext: E H L O L

4. Useful when sender and receiver share the key securely.

Features of Keyed Transposition

Stronger than keyless methods.
Requires sender and receiver to know the key.
Often used in combination with substitution ciphers for extra security.

Q28. What is Stream and Block Ciphers
Stream and block ciphers are types of symmetric-key encryption methods.
1. Stream Cipher
Encrypts one bit or one byte at a time.
Uses a keystream generator to produce a sequence of bits.
Each bit of plaintext is combined with keystream using XOR operation.
Example: RC4, A5/1 (used in GSM).

Advantages:

Fast for real-time data (like voice, video).
Can start encrypting without waiting for full message.

Disadvantages:

Keystream must be truly random and secret.
Vulnerable if keystream repeats.

2. Block Cipher
Encrypts fixed-size blocks of plaintext (e.g., 64-bit, 128-bit).
Each block is encrypted independently using the key.
Example: DES (64-bit block), AES (128-bit block).
Modes of operation: ECB, CBC, CFB, OFB.

Advantages:

Stronger security than stream cipher for bulk data.
Well-suited for files and messages.

Disadvantages:

Slower than stream cipher for real-time data.
Needs padding if plaintext is not exact multiple of block size.

Q29. Basic Working of RSA Algorithm and Its Key Generation RSA is an asymmetric-key encryption algorithm, widely used for secure communication.
Basic Idea

Uses two keys:

Public Key: Used for encryption.
Private Key: Used for decryption.

Based on the difficulty of factoring large prime numbers.

Steps of Key Generation

Choose two large prime numbers: p and q.
Compute n = p × q (n is used in both keys).
Compute φ(n) = (p−1)(q−1) (Euler's totient function).
Choose e such that $1 < e < \varphi(n)$ and $\gcd(e, \varphi(n)) = 1$.
e is the public exponent.

Compute d such that:

$d \times e \mod \phi(n) = 1$

Keys

Public Key: (e, n) → shared with everyone.
Private Key: (d, n) → kept secret by the owner.

Encryption

1. Plaintext message M is converted to a number (M < n).

2. Ciphertext C is calculated using:

$C = M^e \mod n$

Decryption

Ciphertext C is decrypted using private key

$M = C^d \mod n$

Advantages

Strong security due to large primes.
Used for digital signatures and secure data transmission.

Q30. Explain Web Security Threats
Web security threats are risks that can compromise websites, web applications, or online data.
Common Web Security Threats
1. Phishing Attacks

Fake websites or emails trick users into revealing sensitive information like passwords or credit card numbers.

Cross-Site Scripting (XSS)

Attackers inject malicious scripts into webpages viewed by other users.

Can steal cookies or session data.

SQL Injection

Malicious SQL queries are inserted into input fields to access or modify databases.

Malware and Viruses

Web-based malware can infect user devices or servers.

Denial-of-Service (DoS) / Distributed DoS (DDoS)

Overloads the server with traffic, making websites unavailable.

Man-in-the-Middle (MITM) Attack

Attacker intercepts communication between user and server to steal or modify data.

Session Hijacking

Attacker takes control of a valid user session to gain unauthorized access.

Brute Force Attack

Automated attempts to guess passwords or encryption keys.

2. Effects of Web Security Threats

Data theft or loss.
Financial loss.
Loss of user trust.
Website defacement or downtime.

3. Prevention Measures

Use HTTPS and SSL/TLS for secure communication.
Validate user inputs to prevent XSS and SQL injection.
Implement strong passwords and two-factor authentication.
Regularly update software and security patches.

Q31. What are SSL and TLS in Cryptography SSL and TLS are protocols used to secure communication over the Internet.

1. SSL (Secure Sockets Layer)

Developed by Netscape to provide encrypted communication.
Ensures confidentiality, integrity, and authentication.
Works between web browsers and servers.
Uses public-key cryptography for key exchange and symmetric encryption for data transfer.

Versions: SSL 2.0, SSL 3.0 (older and now mostly replaced).

2. TLS (Transport Layer Security)

1. Successor of SSL with improved security.
2. Provides encryption, authentication, and integrity.
3. Versions: TLS 1.0, 1.1, 1.2, 1.3 (latest).
4. Compatible with SSL but more secure against attacks like POODLE.

3. Features of SSL/TLS

1. Encryption: Data is scrambled, preventing eavesdropping.
2. Authentication: Ensures the server is legitimate (via digital certificates).
3. Data Integrity: Detects if data is altered during transmission.
4. Secure Communication: Used for HTTPS, emails, VPNs, and online transactions.

Q32. Difference Between SSL and TLS

SSL and TLS are both protocols for secure communication, but TLS is the improved version of SSL.

1. Security

SSL: Older, less secure. Vulnerable to attacks like POODLE.
TLS: More secure with stronger encryption algorithms and key exchange methods.

2. Versions

SSL: SSL 2.0, SSL 3.0 (mostly obsolete).
TLS: TLS 1.0, 1.1, 1.2, 1.3 (latest and widely used).

3. Protocol Layer

SSL: Works at the transport layer but partially at application layer.
TLS: Strictly works at the transport layer and fully optimized.

4. Cipher Suites

SSL: Supports fewer, weaker cipher suites.
TLS: Supports modern and stronger cipher suites for encryption and authentication.

5. Handshake Process

SSL: Handshake is slower and less efficient.
TLS: Handshake is optimized, faster, and secure.

6. Usage

Both are used for HTTPS, emails, VPNs, and secure transactions, but TLS is recommended today.

Q33. Overview of SSL/TLS Protocol Stack

SSL and TLS provide secure communication over the Internet and sit between the application layer and the transport layer in the network stack.

Layers of SSL/TLS Protocol Stack

1. Application Layer:

Protocols like HTTP, FTP, SMTP, IMAP use SSL/TLS for secure communication.
Provides encryption and authentication to applications.

2. SSL/TLS Record Layer:

Responsible for data fragmentation, compression, encryption, and integrity check.
Divides data into manageable blocks called records.

3. Handshake Protocol:

Used to establish a secure session.

Steps:

1. Client hello → proposes encryption algorithms.
2. Server hello → selects algorithms and sends certificate.
3. Key exchange → session key is generated.
4. Finished → session ready for secure communication.

4. Change Cipher Spec Protocol:

Signals that subsequent data will be encrypted using agreed cipher and keys.

5. Alert Protocol:

Notifies about errors or session termination (e.g., bad certificate, handshake failure).

Features

1. Ensures confidentiality, integrity, and authentication.
2. Can work over any reliable transport protocol like TCP.
3. Supports modern encryption algorithms like AES and SHA.

Q34. Connection Initiation in HTTPS

HTTPS (Hypertext Transfer Protocol Secure) uses SSL/TLS to secure communication between a client (browser) and a server.

Steps of Connection Initiation

1. Client Hello:

Browser (client) sends a “hello” message to the server.
Includes supported SSL/TLS versions, cipher suites, and a random number.

2. Server Hello:

Server responds with chosen SSL/TLS version and cipher suite.
Sends server certificate containing its public key.

3. Certificate Verification:

Client checks if the server certificate is valid and trusted (issued by a Certificate Authority).

4. Key Exchange:

Client and server generate a session key for encrypting data.
Can use RSA, Diffie-Hellman, or other key exchange methods.

5. Finished Messages:

Both client and server send “finished” messages to confirm the session is secure.

6. Secure Session Established:

Browser and server can now exchange encrypted data.

Features of HTTPS Connection

Confidentiality: Data is encrypted.
Integrity: Data is protected from tampering.
Authentication: Server identity is verified.

Q35. Basic Concept of Secure Electronic Transactions (SET)
SET is a protocol designed to secure electronic payments over the Internet, especially for credit card transactions.

1. Purpose of SET

1. Ensure confidentiality, integrity, and authentication of online payments.
2. Prevent credit card fraud during Internet transactions.
3. Enable secure communication between cardholder, merchant, and bank.

2. Components of SET

1. Cardholder: Person making the payment using a credit card.
2. Merchant: Online store or service provider receiving the payment.
3. Acquirer Bank: Bank of the merchant that processes payment.
4. Issuer Bank: Bank that issued the cardholder's credit card.
5. Payment Gateway: Secure network channel for transmitting payment data.

3. Working of SET

1. Cardholder Initiates Transaction:

Sends order information to merchant, encrypted using merchant's public key.

2. Payment Information Sent:

Credit card details are encrypted with bank's public key.

3. Verification:

Merchant sends encrypted payment info to bank for authorization.
Bank verifies the cardholder and approves transaction.

4. Confirmation:

Merchant receives authorization from bank and completes the order.

Cardholder receives confirmation of payment.

4. Features of SET

1. Confidentiality: Card details cannot be seen by merchants.
2. Integrity: Ensures data is not altered during transmission.
3. Authentication: Confirms identity of cardholder and merchant.
4. Uses digital certificates for secure communication.

Advantages

1. Reduces online fraud.
2. Protects sensitive payment information.
3. Ensures trust between buyer, merchant, and bank.

Q36. Difference Between SSL and SET

SSL and SET are both used for secure online transactions, but they work differently.

Purpose

SSL (Secure Sockets Layer):

Provides a secure channel for all types of data over the Internet.
Mainly protects data in transit between client and server.

SET (Secure Electronic Transaction):

Specifically designed to secure credit card payments online.
Protects cardholder and merchant information.

Authentication

SSL:

Authenticates the server to the client.
Client authentication is optional.

SET:

Authenticates both cardholder and merchant using digital certificates.

Confidentiality

SSL:

Encrypts all data between client and server.

SET:

Encrypts order information and payment information separately.

Merchant cannot see credit card details.

Use Cases

SSL:

HTTPS websites, email, VPN, general secure communication.

SET:

Online shopping and credit card transactions.

Security Level

SSL: Good for general secure communication.

SET: Provides higher security for financial transactions.

Q37. What are Intruders? Explain Types of Intruders

Definition of Intruders

An intruder is an unauthorized person who tries to access a computer system or network.
Goal: Steal, modify, or destroy data, or disrupt system operations.

Types of Intruders

(a) Masquerader

Pretends to be an authorized user to gain access.
Uses stolen credentials like username and password.
Goal: Access sensitive data or perform malicious activities.

(b) Misfeasor

An authorized user who misuses their privileges.
May intentionally or unintentionally access unauthorized data.
Example: Employee accessing confidential files without permission.

(c) Clandestine User

Gains unauthorized access without being detected.
Often hides their presence for a long time.
Example: Hacker installing a hidden backdoor in the system.

Q38. How to Protect from Intruders

Protecting a system from intruders involves security measures to prevent unauthorized access, misuse, or attacks.

1. Strong Authentication
Use strong passwords combining letters, numbers, and symbols.
Implement two-factor authentication (2FA) for added security.
Regularly change passwords to prevent misuse.

2. Access Control
Give users only the permissions they need (principle of least privilege).
Limit access to sensitive files and systems.
Use role-based access control (RBAC) or similar methods.

3. Firewalls
Install firewalls to filter incoming and outgoing traffic.
Block suspicious or unauthorized connections.

4. Antivirus and Anti-malware
Use antivirus software to detect and remove malware.
Keep the antivirus up-to-date for latest threats.

5. Intrusion Detection Systems (IDS)
Monitor network or system activities for suspicious behavior.
Alert administrators in case of possible intrusions.

6. System Updates
Regularly update operating systems and software.
Fix security vulnerabilities to prevent intruder attacks.

7. Encryption
Encrypt sensitive data to make it unreadable for intruders.

8. User Awareness
Train users about phishing, suspicious links, and safe practices.
Awareness reduces chances of social engineering attacks.

Q39. Intrusion Detection System (IDS) and Approaches
What is an Intrusion Detection System (IDS)?

IDS is a security system that monitors network or system activity to detect unauthorized access or attacks.
Helps identify intruders before serious damage occurs.
Can alert administrators or take automatic action.

Approaches to Intrusion Detection

(a) Misuse Detection (Signature-Based)

Detects attacks by comparing activity with known attack patterns (signatures).
Works like antivirus software.

Pros:
High accuracy for known attacks.

Cons:
Cannot detect new or unknown attacks.

(b) Anomaly Detection (Behavior-Based)

Detects attacks by monitoring normal system behavior.
Flags activity that deviates from normal patterns as suspicious.

Pros:
Can detect unknown attacks.

Cons:
May give false positives if normal behavior changes.

Features of IDS

Monitors network traffic or system logs.
Alerts administrators of potential intrusions.
Can be network-based (NIDS) or host-based (HIDS).

Q40. What is Firewall? Explain Characteristics of Firewall

What is a Firewall?

A firewall is a network security device or software that controls incoming and outgoing network traffic.
It acts as a barrier between a trusted network (like your LAN) and an untrusted network (like the Internet).
Purpose: Prevent unauthorized access while allowing legitimate communication.

Characteristics of a Firewall

1. Access Control:
Allows or blocks network traffic based on security rules.

2. Monitoring:
Tracks all incoming and outgoing traffic.

3. Logging:
Keeps records of network activity, which helps in detecting attacks.

4. Authentication:
Can require users to verify their identity before granting access.

5. Protection:
Protects network from unauthorized access, hackers, and malware.

6. Traffic Filtering:
Can filter packets based on IP address, protocol, port, or content.

7. Policy Enforcement:
Enforces organizational security policies consistently.

Q41. Explain Needs of Firewalls

Firewalls are important to protect networks and systems from unauthorized access and attacks.

1.Prevent Unauthorized Access

Stops hackers or intruders from entering the network.
Ensures only trusted users can access sensitive systems.

2. Protect Sensitive Data

Safeguards confidential information from being stolen or modified.

Prevents data leakage to external networks.

3. Monitor Network Traffic

Keeps track of incoming and outgoing data.
Helps in detecting suspicious activities or attacks.

4. Enforce Security Policies

Ensures all network users follow organization's security rules.
Can block unwanted services or websites.

5. Reduce Risk of Malware

Can block access to malicious websites.
Prevents viruses, worms, and spyware from entering the network.

6. Support for VPNs

Firewalls can manage VPN connections for secure remote access.
Ensures encrypted communication for remote users.

Q42. Explain All Types of Firewalls

Firewalls can be classified based on functionality, deployment, and filtering method.

1. Packet-Filtering Firewall

Examines each data packet passing through the network.
Allows or blocks packets based on IP address, port number, and protocol.

Pros: Simple and fast.

Cons: Cannot check packet content; less secure against advanced attacks.

2. Stateful Inspection Firewall

Monitors active connections and keeps track of packet states.
Decides whether to allow packets based on connection context.

Pros: More secure than packet-filtering firewall.

Cons: Slower; requires more resources.

3. Proxy Firewall (Application-Level Firewall)

Works at application layer by acting as an intermediary between client and server.
Can inspect actual content of traffic (HTTP, FTP, email).

Pros: Strong security, hides internal network.

Cons: Slower; complex to configure.

4. Next-Generation Firewall (NGFW)

Combines traditional firewall features with intrusion detection/prevention and deep packet inspection.
Can block advanced threats like malware and ransomware.

Pros: Highly secure and modern.

Cons: Expensive; requires skilled administration.

5. Host-Based Firewall

Installed on individual computers.
Monitors incoming and outgoing traffic on that host.

Pros: Protects individual machines.

Cons: Only protects that device; not the whole network.

6. Cloud Firewall

Deployed in the cloud environment to protect cloud applications and data.
Can scale according to network traffic.

Pros: Flexible and scalable.

Cons: Dependent on cloud service provider.

Q43. Advantages and Disadvantages of Firewalls

Firewalls are essential for network security, but they have both pros and cons.

Advantages of Firewalls

Prevent Unauthorized Access:
Stops hackers or intruders from entering the network.

Protect Sensitive Data:
Safeguards confidential information from theft or misuse.

Monitor Network Traffic:
Tracks incoming and outgoing data to detect suspicious activity.

Enforce Security Policies:
Ensures users follow the organization's rules and block unwanted traffic.

Provide VPN Support:
Allows secure remote access for employees.

Reduce Malware Risk:
Blocks malicious websites and harmful data packets.

Customizable Rules:
Admins can configure firewall rules based on network needs.

Disadvantages of Firewalls

Cannot Stop All Threats:
Firewalls cannot protect against internal attacks, phishing, or social engineering.

Performance Impact:
High traffic filtering can slow down network performance.

Complex Configuration:
Advanced firewalls require skilled administration.

False Sense of Security:
Users may rely only on firewall, ignoring other security measures.

Cost:
Next-generation firewalls or cloud firewalls can be expensive.